



7.7.6 Wrap-Up: Reflection

1. Choose the emoji that reflects your understanding of factoring perfect square trinomials.



0 1 2 3 4 5 6 7 8 9 10

2. Complete the statement.

I chose this emoji because _____



Unit 7, Lesson 8: Factoring the Difference of Two Squares



Benchmark

MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real number system.



Learning Targets

- ☐ I can factor a difference of two squares.
- ☐ I can rewrite a polynomial expression as a product of polynomials by factoring.



7.8.1 Warm-Up: Bell Work

1. Factor $36x^2 - 12x + 1$.

2. Factor $9y^2 + 30y + 25$.



7.8.2 Exploration: What Is a Difference of Two Squares?

The table shows two algebra tile models and their corresponding expressions.

$4x^2 - 12x + 9$						$4x^2 - 9$					
x^2	x^2	$-x$	$-x$	$-x$		x^2	x^2	x	x	x	
x^2	x^2	$-x$	$-x$	$-x$		x^2	x^2	x	x	x	
$-x$	$-x$	1	1	1		$-x$	$-x$	-1	-1	-1	
$-x$	$-x$	1	1	1		$-x$	$-x$	-1	-1	-1	
$-x$	$-x$	1	1	1		$-x$	$-x$	-1	-1	-1	

The quadratic expression $4x^2 - 12x + 9$ is a perfect square trinomial.

- Discuss with your partner why the quadratic expression $4x^2 - 9$ is not a perfect square trinomial, even though the algebra tile model is a square. Summarize your discussion.

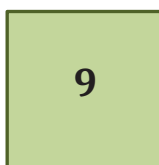
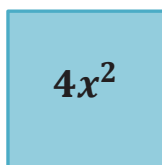
2. Use the algebra tile models to factor each quadratic expression.

$4x^2 - 12x + 9$	$4x^2 - 9$

3. What do you notice about the two factored forms?

4. The quadratic expression $4x^2 - 9$ is a difference of two squares. Discuss with your partner why $4x^2 - 9$ is called a "difference."

5. The two squares shown represent each term of the binomial $4x^2 - 9$.



What is the side length of each square?

$4x^2$	9
Side Length = _____	Side Length = _____

The squares are a geometric representation of each term. The quadratic expression $4x^2 - 9$ can be rewritten with each term as a monomial squared, as shown.

$$(2x)^2 - (3)^2$$

6. Discuss with your partner how the factored form of $4x^2 - 9$ reflects the expression $(2x)^2 - (3)^2$. Summarize your discussion.



The **difference of two squares** is a binomial of the form $(a)^2 - (b)^2$, where a and b can represent a monomial expression or any value except 0. The factored form is $(a + b)(a - b)$.

7. Use the table to determine if each expression is a difference of two squares.

Quadratic Expression	The Expression is a Binomial that Can Be Written as a Difference	First Term is a Perfect Square	Second Term is a Perfect Square
$-100g^4 + 1$	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
$16y^2 - 100y$	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
$-b^2 + 25$	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
$25w^2 - 40w + 16$	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
$9x^2 - 4y^6$	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
$q^2 - 2qz + z^2$	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
$49g^2 - 81h^2$	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No

- Complete the table.
- Circle the expressions in the table that are considered a difference of two squares, based on your check marks.



7.8.3 Guided Practice: Factoring a Difference of Two Squares

1. Consider the expression $49g^2 - 81h^2$.

- Rewrite $49g^2 - 81h^2$ as a difference of two squares.

$$(\quad)^2 - (\quad)^2$$

- Factor $49g^2 - 81h^2$.

2. Consider the expression $9x^2 - 4y^6$.

- Rewrite $9x^2 - 4y^6$ as a difference of two squares.

$$(\quad)^2 - (\quad)^2$$

- Factor $9x^2 - 4y^6$.

3. Both $-121y^2 + 81$ and $-1 + 16x^2$ are differences of two squares. Discuss with your partner how to rewrite them as a difference of two squares. Then, work together to factor each.

$-121y^2 + 81$	$-1 + 16x^2$



7.8.4 Try It: Factoring Polynomials

This is a mixed practice, which includes all of the different types of polynomials factored in this unit. If necessary, review previous lessons for examples of how to factor each type.

1. Factor each expression.

a. $49t^2 - 42t + 9$

b. $1 - 25k^2$

c. $36x^2y^7 - 48x^4y^4 + 12x^2y^4$

d. $12k^2 + 3k - 15$

e. $d^2 + d - 42$



7.8.5 Your Turn!

1. Factor each expression.

a. $25m^2 + 20m + 4$

b. $2g^2 + 3g - 20$

c. $y^2 - 5y - 84$

d. $49z^4 - 16w^4$



7.8.6 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept.



I need help.



I got it.



I could help someone.

Factoring out a common monomial factor			
Factoring a trinomial where $a = 1$			
Factoring a trinomial where $a > 1$			
Factoring a perfect square trinomial			
Factoring a difference of two squares			